

CCNA 200-301 V2.0

# 07 The IOS Command Line

Part I — Foundations

EXAM TOPICS

2.4

LECTURER

**Dr. Mustafa Hameed**



**THE ISLAMIA UNIVERSITY OF BAHAWALPUR**

Department of Information Technology

# Where this chapter sits

---

|                 |                                         |                |
|-----------------|-----------------------------------------|----------------|
| <b>Part I</b>   | <b>Foundations</b>                      | chapters 1–7   |
| <b>Part II</b>  | Switching and Network Access            | chapters 8–14  |
| <b>Part III</b> | IP Routing                              | chapters 15–19 |
| <b>Part IV</b>  | Network Services and Security           | chapters 20–26 |
| <b>Part V</b>   | Wireless, Virtualisation and Operations | chapters 27–32 |
| <b>Part VI</b>  | Sitting the Exam                        | chapters 33–34 |

# What you will be able to do

---

- **Move** between IOS command modes without hesitating, and recognise which mode a prompt indicates.
- **Distinguish** the running configuration from the startup configuration, and save deliberately rather than by habit.
- **Perform** a first-time device setup: hostname, addressing, passwords, SSH and a banner.
- **Filter** `show` output, which is what makes long output usable under exam time pressure.

# Before we start

---

Chapter 4 for addressing and Chapter 6 for the diagnostic commands you will now be typing for real.

# Vocabulary for this chapter

---

user EXEC

privileged EXEC

global configuration

interface configuration

line configuration

running-config

startup-config

NVRAM

context-sensitive help

pipe

banner

SSH

Each is defined where it first matters — not here.

# Getting out, and the two ways to do it

---

`exit` steps back one level. `end`, or `Ctrl-Z`, jumps straight to privileged EXEC from anywhere. When an exam question shows you a prompt, the prompt is telling you which commands are currently legal — a global command typed in interface mode is rejected, and that is a favourite trap.

One shortcut worth knowing: `do` runs a privileged EXEC command without leaving configuration mode, so `do show ip interface brief` works from `SW1(config-if)#`.

# Two configurations, and only one survives a reboot



# running-config and startup-config

---

The **running configuration** is in RAM. It is what the device is doing right now, and every command you type takes effect there *immediately*.

The **startup configuration** is in NVRAM. It is what the device will load when it next boots. Nothing you type reaches it until you copy it there.

- `copy running-config startup-config` — save. Abbreviated `wr` or `write memory` on most platforms.
- `show running-config` — what is in force.
- `show startup-config` — what will be in force after a reload.

# Not saving is a technique, not only a mistake

---

Because configuration takes effect immediately and is lost on reload, a deliberately unsaved change is a safety net. If you are about to modify the access list or the interface you are connected through, and you get it wrong, a power cycle restores the last saved configuration and your access with it.

The professional habit is: make the risky change, verify you still have access, *then* save. Reversing that order is how people lock themselves out of devices in another building.

# Getting help, and typing less



# Filtering output is a survival skill

---

A full `show running-config` on a production switch runs to hundreds of lines. The pipe cuts it to what you need, and in an exam it is the difference between answering and scrolling.

- `| include vlan` — only lines containing “vlan”
- `| exclude !` — drop the `!` comment lines
- `| begin interface` — start output at the first match
- `| section interface Gi0/1` — that interface’s whole block

# show running-config | section interface GigabitEthernet0/1

---

SW1#

```
interface GigabitEthernet0/1
  description Uplink to SW2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30
```

# A first-time setup, end to end



# A switch from the box to remotely manageable

## IOS · configuration mode

```
enable
configure terminal
!
! 1. Identify the device. The hostname appears in every prompt and log.
hostname SW1
!
! 2. Protect privileged mode. "secret" is hashed; "password" is not --
!     never use "enable password".
enable secret Str0ng-Enable-Secret
!
! 3. A local user for SSH, with full privilege.
username netadmin privilege 15 secret Str0ng-User-Secret
!
! 4. A management address. A layer 2 switch has no routed ports, so
!     this lives on an SVI, and it needs a gateway to be reachable
!     from another subnet.
interface Vlan1
  ip address 10.0.0.11 255.255.255.0
  no shutdown
exit
ip default-gateway 10.0.0.1
```

# A switch from the box to remotely manageable (continued)

## IOS · configuration mode

```
!  
! 5. SSH needs a domain name and a key. 2048-bit minimum; version 2 only.  
ip domain-name example.local  
crypto key generate rsa modulus 2048  
ip ssh version 2  
!  
! 6. Console: require login, and stop output interrupting your typing.  
line console 0  
  login local  
  logging synchronous  
  exec-timeout 10 0  
exit  
!  
! 7. Remote access: SSH only, authenticated against local users.  
line vty 0 15  
  login local  
  transport input ssh  
  exec-timeout 10 0  
exit  
!
```

# A switch from the box to remotely manageable (continued)

---

IOS · configuration mode

```
! 8. A legal banner. Never write "welcome" -- it has been argued in court.  
banner motd # Authorised access only. Activity is logged. #  
!  
end  
copy running-config startup-config
```

VERIFY

# show ip ssh

---

```
SW1#
```

```
SSH Enabled - version 2.0
```

```
Authentication timeout: 120 secs; Authentication retries: 3
```

# Common pitfalls

---

- **Using `enable password`.** It is stored in clear text. `enable secret` is hashed, and where both exist the secret wins.
- **Forgetting `transport input ssh`.** Leave it at the default and Telnet stays available, which undoes the point of configuring SSH.
- **Generating a key without a hostname and domain name.** The key is named after them, and the command fails or produces a useless key if they are still at defaults.
- **Saving before verifying.** See the warning above. Verify access, then save.
- **Omitting `logging synchronous`.** Without it a log message arrives in the middle of the line you are typing, and you cannot tell what you have actually entered.

## The change worked yesterday and the device came back without it.

A colleague configured a new VLAN and confirmed it working. After a power cut overnight, the VLAN is gone and the configuration looks like it did last week.

**The change worked yesterday and the device came back without it.**

**What would you check first — and what do you expect to see?**

## show archive config differences nvram:startup-config system:running-config

SW1#

!Contextual Config Diffs:

+vlan 30

+ name Guest

+interface GigabitEthernet0/9

+ switchport access vlan 30

**What is the fault — and what does this output rule out?**

## Why it is doing that

**Diagnosis.** The change was made in the running configuration and never copied to the startup configuration. The reload loaded NVRAM, which did not have it. Nothing is broken — the device did exactly what it was told.

**Confirm it** by comparing the two before doing anything else:

Any output at all means the two differ and a reload will lose the difference.

## And the lesson

**Fix.** Re-apply the change, verify, then `copy running-config startup-config`. As a habit, run that comparison before you leave any device.

# Bring up a switch you can reach from your desk

---

**Topology.** One switch, one PC on the same VLAN, connected by console cable and by an Ethernet cable.

1. Connect by console. Move through all five modes in turn and write down the prompt at each. Use `end` to jump back from the deepest.
2. Set hostname, `enable secret` and a local user.
3. Give `Vlan1` an address in the PC's subnet and a default gateway. Ping the switch from the PC to prove it.
4. Enable SSH properly: domain name, 2048-bit key, version 2, and `transport input ssh` on the VTY lines.
5. SSH from the PC. Then try Telnet and confirm it is refused.

## Bring up a switch you can reach from your desk (continued)

---

6. Make a small change, do *not* save, reload, and observe the change vanish. Repeat with a save.
7. **Extension.** Practise `| include`, `| section` and `| begin` on `show running-config` until you can extract a single interface block in one command without thinking.

CHECKPOINT 1 of 3

You are at `SW1(config-if)#` and need to check the routing table without losing your place. What do you type?

Discuss · then advance

## Checkpoint 1 of 3

---

**You are at SW1(config-if)# and need to check the routing table without losing your place. What do you type?**

`do show ip route`. The `do` prefix runs a privileged EXEC command from any configuration mode without leaving it.

CHECKPOINT 2 of 3

Which configuration does a device load on boot, and which one are you editing when you type a command?

Discuss · then advance

## Checkpoint 2 of 3

---

**Which configuration does a device load on boot, and which one are you editing when you type a command?**

It boots from `startup-config` in NVRAM. You are editing `running-config` in RAM. Anything not saved is lost at the next reload — see Chapter 31 for how this catches out automation too.

CHECKPOINT 3 of 3

Why must a hostname and domain name be set before generating an RSA key?

Discuss · then advance

## Checkpoint 3 of 3

---

### Why must a hostname and domain name be set before generating an RSA key?

Because the RSA key pair is named after the fully qualified domain name, `hostname.domain`. Without both, the device has no name to label the key with and refuses to generate it.

# What to take away

---

- The prompt tells you the mode, and the mode decides which commands are legal. `end` returns to privileged EXEC from anywhere; `do` runs one EXEC command without leaving.
- Commands take effect in the running configuration immediately and are lost on reload until copied to the startup configuration.
- Verify access first, save second. An unsaved risky change is a safety net.
- `enable secret`, not `enable password`. SSH version 2 with a 2048-bit key, and `transport input ssh` to shut off Telnet.
- Filtering with `| include`, `| section` and `| begin` is what makes device output usable at speed.

# Where we got to

- The prompt tells you the mode, and the mode decides which commands are legal. `end` returns to privileged EXEC from anywhere; `do` runs one EXEC command without leaving.
- Commands take effect in the running configuration immediately and are lost on reload until copied to the startup configuration.
- Verify access first, save second. An unsaved risky change is a safety net.
- `enable secret`, not `enable password`. SSH version 2 with a 2048-bit key, and `transport input ssh` to shut off Telnet.

NEXT

**Chapter 8 — VLANs and Access Ports**